

Project Design Phase
Problem – Solution Fit Template

Date	04 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Problem–Solution Fit Statement

Problem:

People living in flood-prone areas struggle to receive timely and accurate flood warnings, resulting in delayed evacuations, property damage, financial losses, and risks to human life.

Solution:

Rising Waters leverages machine learning to analyse historical and real-time weather, rainfall, river-level, and environmental data to predict floods accurately. The system provides location-specific alerts, interactive flood-risk maps, and emergency notifications, enabling users and authorities to take preventive action before flooding occurs.

Problem–Solution Fit:

The solution directly addresses customers' need for **early, accurate, and location-based flood predictions**, improving disaster preparedness, minimizing losses, and enhancing community safety. It integrates seamlessly into users' existing behaviour by delivering alerts through web and mobile applications, making adoption simple and effective.

Problem–Solution Fit Template

Section	Details
Target Customer	Residents living in flood-prone areas, Farmers, Disaster Management Authorities, Local Government Agencies, Emergency Response Teams
Customer Problem	Existing flood warning systems often provide delayed or inaccurate alerts. People do not receive location-specific flood predictions, making it difficult to prepare for floods, protect property, and evacuate safely.
Current Situation	Customers rely on weather forecasts, television, radio, social media, and local announcements, which may not provide accurate or timely flood predictions. Manual monitoring of rainfall and river levels is time-consuming and unreliable.
Customer Behaviour	Users frequently check weather apps, follow news updates, monitor river water levels, contact local authorities, and depend on community warnings before taking action.
Pain Points	• Late flood warnings • Inaccurate predictions • Property and crop damage • Financial losses • Limited evacuation time • Lack of real-time location-based alerts

Section	Details
Proposed Solution	Develop a Machine Learning-based Flood Prediction System that analyzes historical rainfall, weather forecasts, river water levels, and environmental data to predict flood risks accurately and send real-time alerts to users.
How the Solution Solves the Problem	• Provides early flood predictions using machine learning algorithms. • Sends real-time notifications based on the user's location. • Displays flood risk levels on interactive maps. • Assists disaster management authorities in planning evacuation and emergency response.
Customer Benefits	• Early warning before floods occur. • Improved family and community safety. • Reduced property and crop damage. • Better preparedness and faster emergency response. • Increased confidence in flood prediction accuracy.
Behaviour Change	Instead of waiting for manual weather updates, users receive instant AI-powered flood alerts and can make proactive decisions, such as preparing emergency kits, protecting valuables, or evacuating early.
Value Proposition	"Accurate, real-time flood prediction powered by Machine Learning to protect lives, property, and livelihoods through timely alerts and data-driven decision-making."
Success Metrics	• Increased prediction accuracy (>90%) • Reduced false flood alerts • Faster alert delivery (<1 minute) • Increased user adoption • Reduced flood-related losses and improved emergency response time

Define CS, fill the CS	1. CUSTOMER SEGMENT(S) CS Who is your customer? i.e. residents, farmers, disaster authorities, NGOs, local government, businesses	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices or solutions? i.e. lack of accurate data, high cost, limited internet, low awareness, unreliable devices	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem? i.e. weather apps, TV/radio alerts, manual monitoring, government alerts, social media updates	Explore AS, differentiate
	- Residents in flood-prone areas - Farmers and rural communities - Disaster Management Authorities - Local Government / Urban Planners - Emergency Response Teams - NGOs and Relief Organizations	- Inaccurate or delayed flood warnings - Lack of real-time, location-specific information - Limited access to reliable data and technology - Low awareness about flood preparedness - High cost of existing flood monitoring systems - Poor communication and coordination during emergencies	- Weather forecast apps and websites - TV, radio and newspaper alerts - Government and local authority announcements - Social media and community groups - Manual monitoring of rainfall and river levels	
Focus on AP to find RC, understand RC	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (problems) do you address for your customers? There could be more than one, explore different jobs.	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the basic faulty behind the need to do this job? i.e. customer lacks access to real-time data and predictive insights	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? i.e. check weather apps, call authorities, monitor news, ask neighbors, avoid risky areas	Focus on JB, top two BE, understand RC
	- Get early and accurate flood warnings - Monitor real-time weather and river levels - Prepare and protect lives, property and crops - Make data-driven decisions for evacuation - Coordinate relief and emergency operations	- Lack of real-time data integration - Inaccurate prediction models - Fragmented information across multiple sources - Manual data collection and monitoring - Delayed communication of warnings	- Check weather forecast and news updates - Call local authorities or community leaders - Monitor river water levels manually - Share information on social media groups - Evacuate only after severe warnings	
Identify triggers TR & EM	3. TRIGGERS TR What triggers customers to act? i.e. seeing dark clouds, heavy rainfall, warnings on TV, rising river levels	10. YOUR SOLUTION SL If you are working on an existing solution, write down your current solution first. If in the canvas, and check how much it fits into your existing one. Write down your new solution (better than step 1 blank until you fill in the canvas) and compare it with the customer limitations, where it aligns and matches customer behaviour.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7	Extract online & offline CH of BE
	- Heavy rainfall or continuous rain - Rising river or dam water levels - Weather alerts and advisories - Past flood experiences - Government notifications	Rising Waters: A Machine Learning Approach to Flood Prediction - Real-time data collection from weather APIs, sensors, and satellites - Machine learning models predict flood risk accurately - Location-based alerts and early warnings - Interactive flood risk maps and dashboards - Timely notifications to residents and authorities - Actionable insights for preparedness and response	- Mobile App & Web Dashboard - Social Media Platforms - Email & SMS Alerts	
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? i.e. fear, stress, anxiety before; relief, safety, confidence after		8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.	
	Before: Fear, Uncertainty, Anxiety, Stress After: Relief, Safety, Confidence, Preparedness		- Local community meetings - TV / Radio Announcements - Sirens / Public Address Systems	

References:

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>